

Expt 3	Performance Analysis of IIR Filters
Date	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

Code:

```

clc;
clear;
close all;

%-----
% EX NO : 3
% TITLE : Comparison of Butterworth and Chebyshev
%         Type-I Band-Pass IIR Filters Using MATLAB
%-----

% Filter Specifications
Fs = 2000;      % Sampling Frequency (Hz)
Rp = 1;         % Passband Ripple (dB)
Rs = 40;        % Stopband Attenuation (dB)

Wp = [300 600] / (Fs/2); % Passband Edge Frequencies
Ws = [200 700] / (Fs/2); % Stopband Edge Frequencies

%-----
% Butterworth Band-Pass Filter Design
%-----
[N1, Wn1] = buttord(Wp, Ws, Rp, Rs);
[b1, a1] = butter(N1, Wn1, 'bandpass');

%-----
% Chebyshev Type-I Band-Pass Filter Design
%-----
[N2, Wn2] = cheb1ord(Wp, Ws, Rp, Rs);
[b2, a2] = cheby1(N2, Rp, Wn2, 'bandpass');

%-----
% Stability Check
%-----
if all(abs(roots(a1)) < 1)
    butterStable = 'STABLE';
else
    butterStable = 'UNSTABLE';
end

if all(abs(roots(a2)) < 1)

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        chebyStable = 'STABLE';
else
    chebyStable = 'UNSTABLE';
end

%-----
% Frequency Response
%-----
figure;
freqz(b1, a1, 1024, Fs);
title('Butterworth Band-Pass Filter');

figure;
freqz(b2, a2, 1024, Fs);
title('Chebyshev Type-I Band-Pass Filter');

%-----
% Group Delay Comparison
%-----
figure;
grpdelay(b1, a1);
hold on;
grpdelay(b2, a2);
grid on;
legend('Butterworth', 'Chebyshev Type-I');
title('Group Delay Comparison');

%-----
% Pole-Zero Plots
%-----
figure;

subplot(1,2,1);
zplane(b1, a1);
title('Butterworth Pole-Zero Plot');

subplot(1,2,2);
zplane(b2, a2);
title('Chebyshev Type-I Pole-Zero Plot');

%-----
% Impulse Responses
%-----
figure;

subplot(2,1,1);
impz(b1, a1);
title('Butterworth Impulse Response');

subplot(2,1,2);

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impz(b2, a2);
title('Chebyshev Type-I Impulse Response');

%-----
% Pole Magnitudes
%-----
figure;

subplot(1,2,1);
stem(abs(roots(a1)), 'filled');
grid on;
title('Butterworth Pole Magnitudes');
xlabel('Pole Index');
ylabel('Magnitude');

subplot(1,2,2);
stem(abs(roots(a2)), 'filled');
grid on;
title('Chebyshev Type-I Pole Magnitudes');
xlabel('Pole Index');
ylabel('Magnitude');

%-----
% Display Results
%-----
fprintf('-----\n');
fprintf('Band-Pass IIR Filter Comparison\n');
fprintf('-----\n');

fprintf('Butterworth Filter Order    = %d\n', N1);
fprintf('Butterworth Stability          = %s\n', butterStable);

fprintf('\n');

fprintf('Chebyshev Type-I Filter Order = %d\n', N2);
fprintf('Chebyshev Stability           = %s\n', chebyStable);

fprintf('-----\n');

```

Output:

Fig 1:

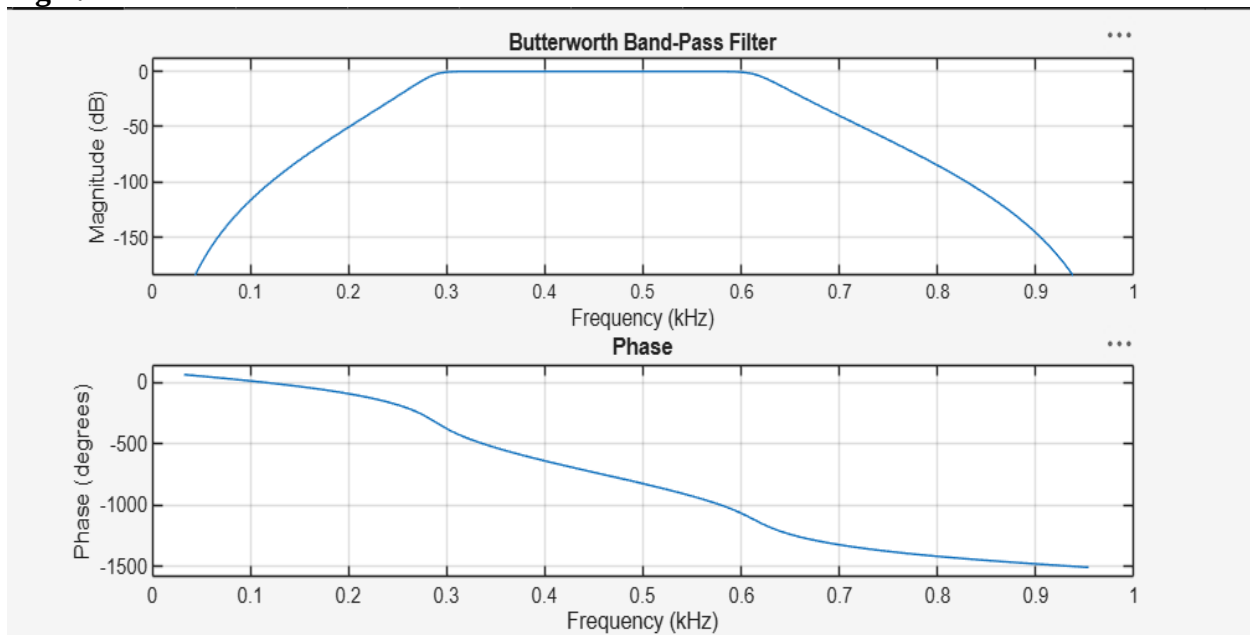


Fig 2:

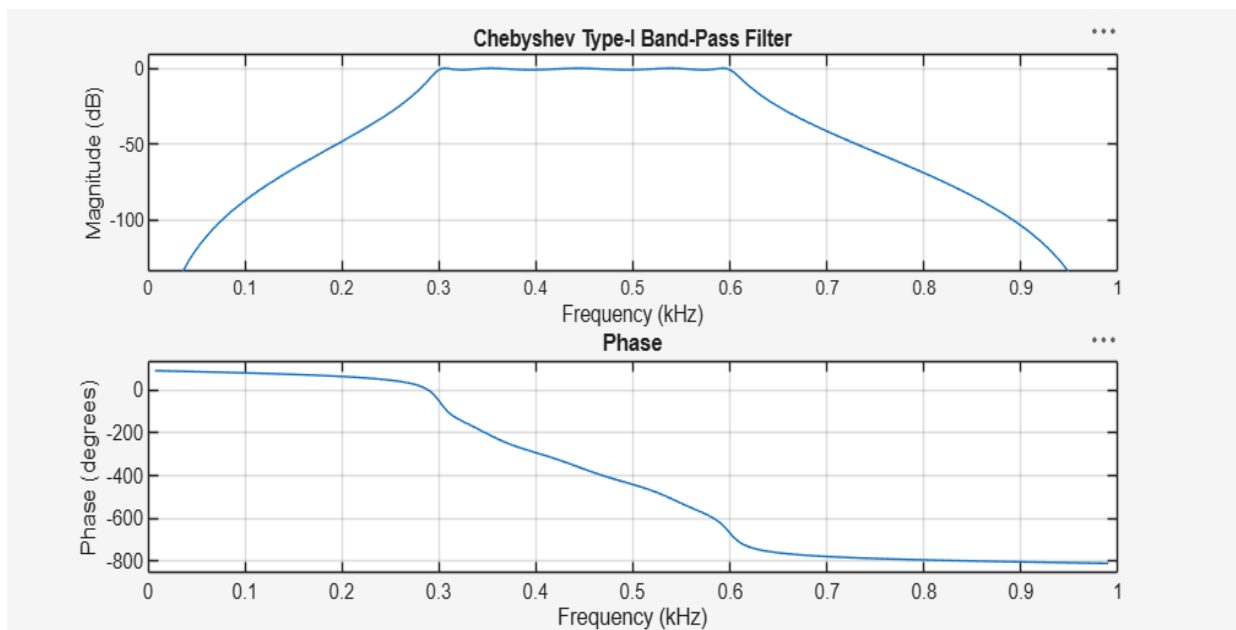


Fig 3:

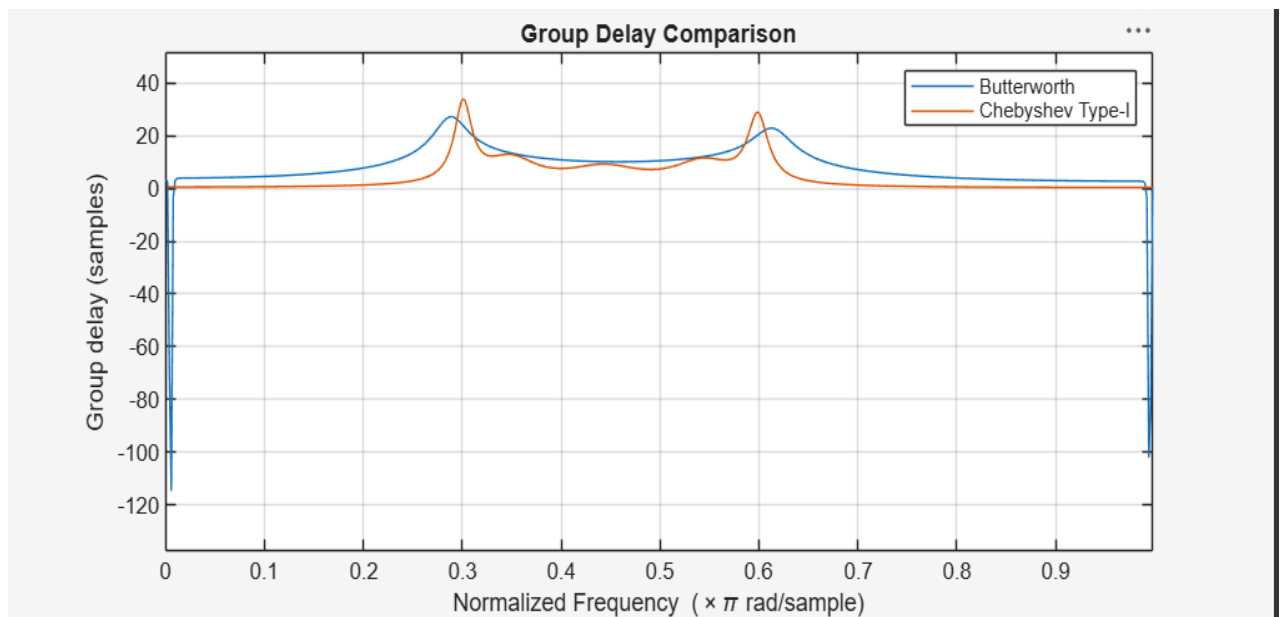


Fig 4:

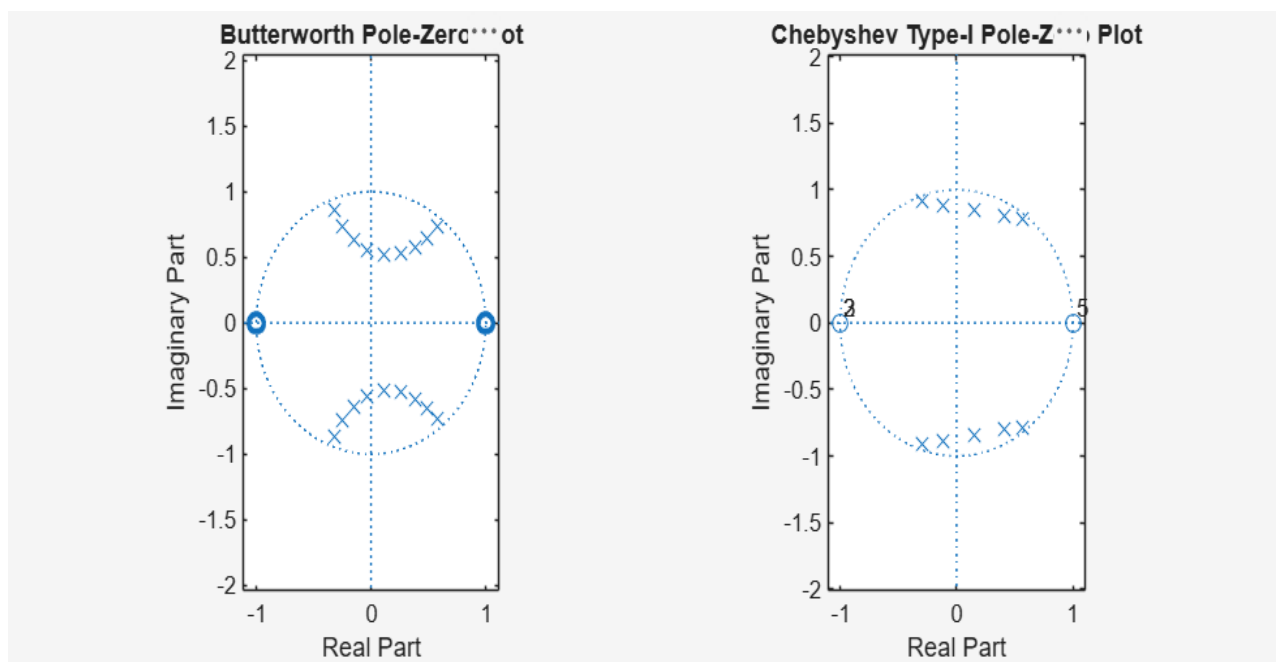


Fig 5:

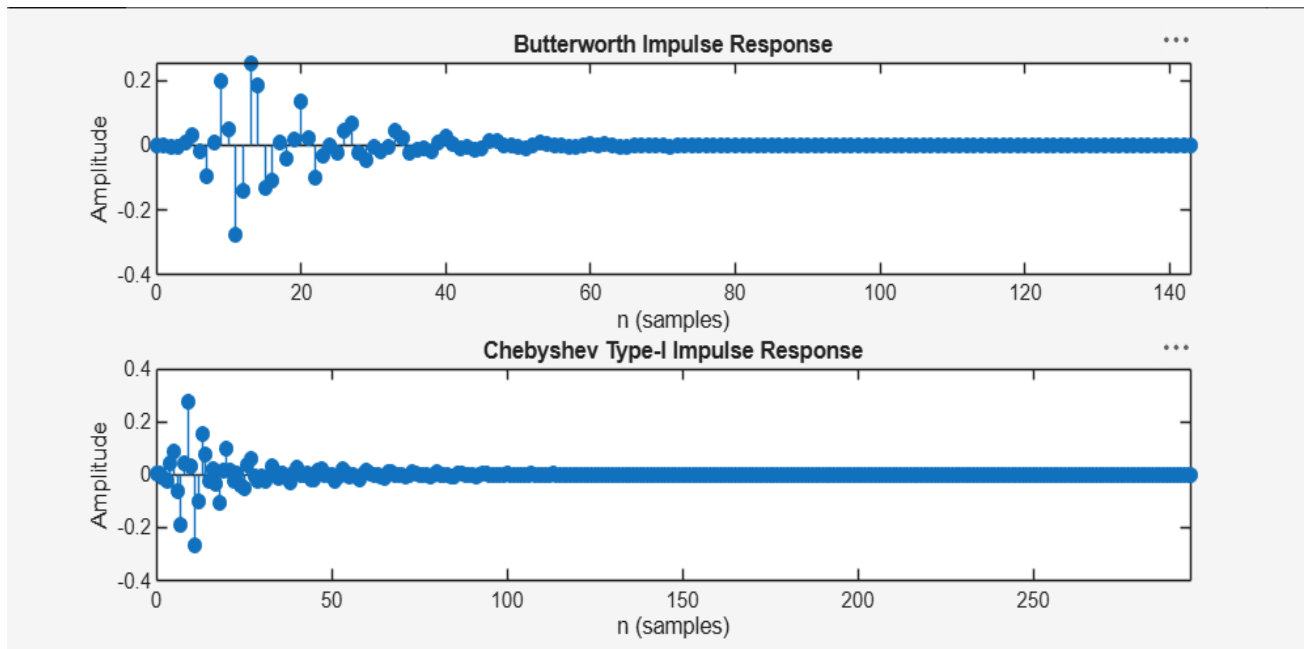


Fig 6:

